

Seamless Spatial Extension in 3D Gaussian Splatting via Auxiliary Wild Views

Chaewon Lee^{1*}, Seunggyu Choi^{1*}, Suwoong Yeom^{1*},
Joonsoo Kim², Kugjin Yun², and Suk-Ju Kang^{1†}

¹ Sogang University

² Electronics and Telecommunications Research Institute

* Indicates Equal Contribution † Corresponding Author

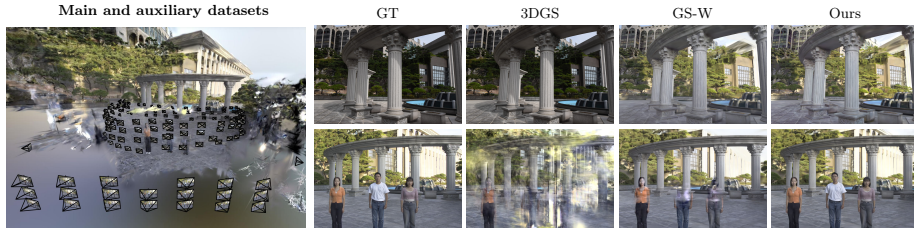


Fig. 1: Given main and auxiliary datasets, our framework successfully expands the scene while preserving core objects and seamlessly integrating color and illumination discrepancies.

Abstract. High-fidelity 3D reconstruction of large-scale stages is essential for immersive VR/AR experiences, yet limited camera deployment restricts spatial coverage. A practical remedy is to expand the scene with auxiliary footage of the surrounding environment captured at a different time, but this introduces severe domain gaps in illumination and scene composition, as core subjects are absent from the auxiliary views. Existing in-the-wild 3D Gaussian Splatting methods fail in this scenario, since they treat such inconsistent subjects as transient noise and eliminate them. We introduce a Context-Decoupled Spatial Extension Framework that factorizes the optimization of the shared static environment and the core subjects. By combining semantic-guided optimization with an asymmetric geometry preservation mechanism, our approach expands the spatial boundaries of the scene while faithfully preserving the geometry and fine details of the core subjects.

Keywords: 3D Gaussian Splatting · 3D Reconstruction · Spatial Extension · In-the-Wild Reconstruction

1 Introduction

With the rapid advancement of virtual reality (VR), augmented reality (AR), and metaverse technologies, there is a surging demand for immersive content that reconstructs vibrant cultural performances and large-scale stages in 3D. Novel view synthesis (NVS) has demonstrated high-quality reconstruction performance with the emergence of NeRF [1, 2, 14, 26] and 3D Gaussian Splatting (3DGS) [9, 10, 22, 24]. However, reconstructing a complete 3D space in a real-world, large-scale stage environment remains a challenging task. In practice, there are strict physical and financial limitations to deploying an unlimited number of cameras on-site. Consequently, a practical setup utilizes a limited number of fixed multi-view cameras to intensively capture key subjects on the main stage during the performance, accompanied by auxiliary footage captured via a hand-held camera during non-performance hours to cover the surrounding wide environment, thereby maximizing the sense of realism and liveliness. Ultimately, a scenario is fundamentally required where the entire space is reconstructed by expanding the surrounding regions using the auxiliary dataset, while preserving the core objects captured by the main dataset.

Integrating the main and auxiliary datasets to reconstruct the entire space introduces complex domain gaps that must be carefully addressed. Because the two sets of footage are captured at different times (during the performance vs. non-performance hours) using different camera devices, they inevitably exhibit severe color and illumination discrepancies. Concurrently, there is a distinct structural inconsistency regarding the composition of the scenes: the auxiliary footage captures an empty stage, whereas the main footage contains core subjects (e.g., performing actors) that must be strictly preserved.

To overcome limited fields of view and extend the space, approaches utilizing the inpainting techniques of generative models, such as WonderJourney [5, 7, 23], have been proposed. However, these generative-based models merely hallucinate visually plausible virtual scenes based on given visual cues. Because they do not physically reconstruct the authentic geometric and visual information of the actual performance environment, they are fundamentally limited for reconstruction tasks that require preserving the true context of the real-world space.

Meanwhile, to register unconstrained images with varying environmental and visual conditions into a consistent scene, various approaches have been widely explored, which can also be utilized as a technique for spatial extension. Pioneering NeRF-based studies [4, 13, 21] introduced the concept of utilizing latent embeddings to separate illumination variations and transient elements from the static background. Building upon this foundation, most recent Gaussian Splatting methodologies [6, 12, 18, 25] have adopted a similar disentanglement strategy by embedding additional parameters into the Gaussians. Through this approach, existing frameworks have successfully integrated appearance discrepancies across varying datasets while significantly improving rendering speed.

However, directly applying these integration models to a scenario for spatial extension presents a structural limitation. These models are inherently designed to treat inconsistent subjects across images as "transient noise", optimizing the

scene reconstruction to completely eliminate them. Consequently, this optimization inevitably leads to the erasure of the core subjects captured in the main footage, rendering such approaches unsuitable for the objective of expanding the surrounding physical space while preserving the structural integrity of the subjects.

To fundamentally address these challenges and resolve the inherent trade-off between environmental alignment and structural preservation, this paper introduces a Context-Decoupled Spatial Extension Framework. Unlike conventional in-the-wild methodologies that unify scene optimization within a single closed loop—often leading to the destruction of target subjects—our framework structurally factorizes the optimization pathways of the shared environment and the core subjects. To seamlessly synchronize disparate datasets without geometric corruption, we leverage a semantic-guided optimization pipeline. This allows us to harmonize cross-domain environmental conditions into a unified static background, while strictly isolating and protecting them from the background optimization pressure. Consequently, our framework effectively expands the spatial boundaries of the original scene using auxiliary datasets, while perfectly preserving the geometric integrity and fine details of the essential target elements.

Our main contributions and the core technical components of the proposed framework can be summarized as follows:

- Context-Decoupled Optimization Framework: We propose a factorized training strategy that structurally decouples the optimization of the shared static background from the intensive reconstruction of the core subjects. By independently optimizing these elements and subsequently integrating them, we successfully prevent the background optimization pressure from degrading the main subjects.
- Semantic mask-based regional and class learning: By leveraging a semantic mask scaled to the range of $[0, 1]$, we localize the loss computation strictly within the targeted region of each phase, while concurrently training the 3D Gaussians to independently learn their own semantic classes (static vs. dynamic).
- Asymmetric geometry preservation mechanism: To robustly fuse datasets with structural inconsistencies, we introduce an asymmetric protection strategy. When training on the auxiliary dataset where the core subjects are absent, we preserve the geometric integrity of the main subjects by temporarily neutralizing the target Gaussians (e.g., opacity-zero rendering) and explicitly freezing their structural parameters (preventing densification and pruning).

2 Related Work

2.1 3D Scene Reconstruction and View Expansion.

The field of Novel View Synthesis (NVS) has achieved remarkable progress since the advent of NeRF, and the recent introduction of 3D Gaussian Splatting (3DGS) has enabled both real-time rendering speeds and outstanding vi-

sual quality. Building upon these 3D reconstruction technologies, various spatial extension studies have been conducted to expand limited capture fields of view. Notably, several approaches, including WonderJourney [23], leverage the inpainting techniques of Diffusion models to extrapolate out-of-view regions. However, these generative-based models merely hallucinate visually plausible virtual scenes based on given image cues. Because they fail to accurately reflect the true geometric information and visual reality of the physical environment where the performance took place, a fundamental limitation exists when the objective is to authentically preserve and spatially extend the real-world context of a stage environment. Meanwhile, active research has also addressed inconsistent observations within a single scene by separating distractors from the background. Robust-3DGS [16] introduces a novel loss for stable novel view synthesis in the presence of distractors, while DeSplat [17] and RobustSplat [8] explicitly separate static and transient Gaussians in a manner similar to our decouple-then-reconstruct strategy.

2.2 Scene Integration in the Wild.

In-the-wild methods integrate images captured at different times or with different devices by modeling appearance variations and separating transient content from static geometry. NeRF-W [13] introduced appearance and transient embeddings, while later 3DGS approaches, including GS-W [25], Wild-GS [19], and WildGaussians [12], adopted related representations to handle illumination and color inconsistencies with efficient rendering.

However, directly applying these integration models to a spatial extension scenario presents a structural limitation. Existing in-the-wild models are inherently designed to treat subjects or elements that are not cross-verified across multiple images as "transient noise," explicitly optimizing the scene reconstruction to eliminate them. Consequently, the core subjects in the main footage, which are absent in the auxiliary footage, are misidentified as noise and erased. Therefore, conventional in-the-wild methodologies cannot be directly applied to a scenario that aims to expand the surrounding physical space using auxiliary data while strictly preserving the structural integrity of the essential core subjects captured in the main data.

3 Method

The overarching objective of the proposed framework is to effectively expand the spatial boundaries of the original scene using auxiliary datasets, while strictly preserving the core objects captured by the main dataset. The overall pipeline of our proposed framework is illustrated in main figure. To achieve aforementioned goals, we first baseline our model on an appearance transfer mechanism of GS-W in section 3.1 and introduce a factorized training strategy that factorizes the optimization process into distinct stages in section 3.2. We leverage a semantic-guided optimization pipeline with soft semantic masks in section 3.3. Addition-

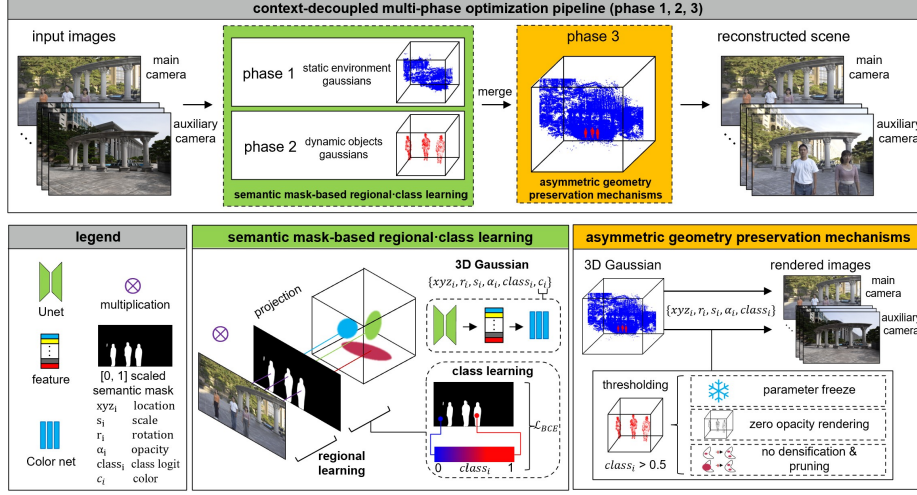


Fig. 2: Overview of our proposed framework. We process an input image containing a mixture of main and auxiliary datasets through a three-phase pipeline. In Phases 1 and 2, semantic mask-based region and class learning are applied, where 3D Gaussians learn class probabilities and losses are computed specifically for each phase’s target region. After each phase, Gaussians corresponding to target regions are extracted via class thresholding. In Phase 3, the extracted Gaussians are merged and optimized using an asymmetric preservation mechanism. For the auxiliary scene, subject Gaussians are filtered by class thresholding, frozen in parameters, rendered with zero opacity to exclude them visually from loss calculation, and protected from densification and pruning. Consequently, the entire scene is reconstructed with unified color tone while preserving the core subject.

ally, we introduce an asymmetric geometry preservation mechanism during the integrated optimization stage in section 3.4.

3.1 Preliminaries

3D Gaussian Splatting (3DGS) 3DGS leverages multi-view 2D images or video data to transform complex 3D spaces into numerous Gaussians, achieving real-time and explicit representation of realistic 3D scenes. A scene is represented as a collection of 3D Gaussian points, where each Gaussian is defined by a central position $\mu \in \mathbb{R}^3$, an opacity $\alpha \in [0, 1]$, and a 3D covariance matrix Σ .

To project the 3D Gaussians onto a 2D image plane, the projected covariance matrix Σ' is derived using the viewing transformation matrix W and the Jacobian of the affine approximation of the projective transformation J as follows:

$$\Sigma' = JW\Sigma W^T J^T \quad (1)$$

For N Gaussians ordered within a specific tile, the final pixel color C is rendered via α -blending:

$$C = \sum_{i=1}^N c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (2)$$

where c_i denotes the color computed through Spherical Harmonics (SH) coefficients and the view direction, and α_i represents the opacity of the i -th Gaussian.

Gaussian in the Wild (GS-W) GS-W, which serves as the foundation of our work, is a framework that extends 3DGS to handle severe illumination and weather variations in unconstrained, in-the-wild environments. To cleanly disentangle the intrinsic scene geometry from transient environmental changes, GS-W extracts a projection feature f_i^P and models sampled features $f_i^1, \dots, f_i^K$ using a U-Net encoder and learnable Gaussian embedding parameters sc_i :

$$(f_i^1, \dots, f_i^K) = BL((sc_i^1, \dots, sc_i^K), (F^1, \dots, F^K)) \quad (3)$$

The dynamic feature $df_i = f_i^P \oplus f_i^1 \oplus \dots \oplus f_i^K$ captures image-specific illumination and environmental contexts. A fusion network integrates df_i with the Gaussian’s intrinsic feature sf_i to form a unified appearance feature af_i . Finally, a color network predicts the color c_i using af_i and view direction ϕ as inputs, replacing conventional SH coefficients:

$$c_i = M_c(af_i, \phi) \quad (4)$$

3.2 Context-Decoupled Multi-Phase Optimization Pipeline

To prevent geometric corruption arising from the asymmetric information gaps between the main and auxiliary datasets, we structure the training process into a multi-phase optimization pipeline. Unlike conventional in-the-wild methodologies that unify scene optimization of the shared environment and transient objects within a single closed loop which often leads to the destruction of target subjects, our framework structurally factorizes the optimization pathways into three distinct stages: Phase 1, Phase 2, and Phase 3. This factorization prevents the optimization pressure from spatial background extension from contaminating the parameter space of the core subjects, thereby successfully securing geometric integrity in the final fused scene.

Prior to optimization, the position μ and color of the 3D Gaussians are initialized based on a sparse point cloud reconstructed by applying a Structure-from-Motion (SfM) [15] algorithm to all images across both the main and auxiliary datasets.

In the initial stages of the pipeline, namely Phase 1 (Shared Static Background Learning) and Phase 2 (Intensive Core Subject Reconstruction), the static environment and the dynamic subjects are trained independently within isolated optimization pathways. To prevent out-of-target environmental noise from infiltrating the Gaussian parameters, we leverage predefined semantic masks

for each input image to localize the loss computation strictly within the targeted region of each phase.

Specifically, Phase 1 utilizes both the main and auxiliary datasets, applying the static region mask $1 - M$ to intensively optimize only the static background Gaussians. This establishes a unified 3D spatial foundation shared across both datasets. Conversely, Phase 2 selectively processes the main dataset where the core subjects are present, applying the dynamic region mask M to independently reconstruct the essential target elements. In a single-loop training architecture, gradients aimed at sharpening background textures often conflict with gradients trying to maintain the dynamic subjects at identical spatial locations, resulting in rendering artifacts. In contrast, our factorized training strategy allows each distinct set of Gaussians to establish its own structural details without mutual gradient interference.

Finally, in Phase 3 (Unified Scene Optimization), the static background Gaussians and dynamic subject Gaussians independently reconstructed in the previous phases are selectively extracted based on their learned semantic classes (Section 3.3) and merged into a single unified space for final optimization. This phase optimizes the entire scene to ensure that the two factorized components blend seamlessly both visually and structurally. Concurrently, it applies an asymmetric preservation mechanism that selectively protects the geometry of the dynamic subjects depending on the source domain of the input view (i.e., whether it originates from the main or auxiliary dataset). Consequently, through the integrated optimization in Phase 3, our framework effectively expands the spatial boundaries using the auxiliary dataset while perfectly preserving the fine details of the core subjects captured in the main dataset.

3.3 Semantic Mask-Based Regional · Class Learning

Semantic Mask-Based Target Region Extraction To optimize the static background and dynamic subjects independently without mutual interference during each phase, our framework leverages semantic masks from SAM3 [3] to isolate the targeted regions. The semantic masks provided with the input images are preprocessed and scaled to a continuous range of $[0, 1]$, where the static background regions correspond to 0 and the dynamic subject regions correspond to 1. This normalized mask is applied to both the input and rendered images via element-wise multiplication. This preprocessing eliminates irrelevant out-of-target environmental noise for the current training phase, thereby localizing the loss computation strictly within the targeted regions during the subsequent L_1 and structural dissimilarity (D-SSIM) loss evaluations.

Semantic Mask-Based Class Learning When proceeding to the integrated optimization in Phase 3 using the Gaussians independently trained during Phase 1 and Phase 2, we enforce individual Gaussians to learn their own semantic class attributes, allowing them to interface seamlessly with the subsequent geometry preservation mechanism. To this end, we introduce a learnable scalar parameter $class_i \in [0, 1]$ into each 3D Gaussian to represent its class probability.

The initial class attributes are optimized independently within their respective target regions during Phase 1 and Phase 2. Once training for these phases concludes, we selectively extract only the target class Gaussians required for each component. This strategy prevents boundary artifacts and misclassification issues that typically arise in a single closed-loop optimization.

During the rendering process, the Gaussians projected onto the current camera viewpoint are extracted and mapped onto the scaled semantic mask plane. Since an individual Gaussian does not map one-to-one with the discrete pixel grid, we utilize the grid sample function to sample continuous semantic mask values based on the 2D projected center coordinates of each Gaussian.

Each Gaussian updates its class attribute $class_i$ using the sampled mask value at its projected location (approaching 0 for static regions and 1 for dynamic regions) as the ground truth. Rather than employing hard values (0 or 1) as in standard binary classification, representing the classes as continuous probabilities minimizes the potential for misclassification at the object boundaries or during early optimization stages, inducing a flexible attribute transition.

Optimization The total loss function designed to achieve regional and class factorization comprises a mask-based reconstruction loss, a semantic class loss, and a depth regularization loss to ensure geometric stability.

The mask-based image loss L_{img}^{mask} restricts reconstruction solely to the targeted region and is computed as a weighted sum of L_1 and D-SSIM losses between the masked rendered image $\hat{I}_m$ and ground-truth I_m :

$$L_{img}^{mask} = (1 - \lambda_D)L_1(\hat{I}_m, I_m) + \lambda_DL_{DSSIM}(\hat{I}_m, I_m) \quad (5)$$

The class loss L_{class} guides projected Gaussians to learn static or dynamic attributes using Binary Cross Entropy between each Gaussian’s predicted class and the semantic mask value at its projected coordinate:

$$L_{class} = -\frac{1}{N} \sum_{i=1}^N [m_i \log(class_i) + (1 - m_i) \log(1 - class_i)] \quad (6)$$

where N denotes the total number of Gaussians projected onto the view, $class_i$ is the predicted class of the i -th Gaussian, and m_i is the corresponding mask value.

Including depth regularization L_{depth} [20] to prevent inaccurate depth placement, the total loss function L_{total} is defined as:

$$L_{total} = L_{img}^{mask} + \lambda_1 L_{class} + \lambda_2 L_{depth} \quad (7)$$

We set the loss weights to $\lambda_1 = 0.01$ and $\lambda_D = 0.2$, while λ_2 is exponentially decayed from 1.0 to 0.01 throughout training.

In Phase 3 (Unified Scene Optimization), L_{class} is omitted as class extraction is complete. To optimize full-scene fidelity without regional constraints, the semantic mask is removed and standard reconstruction loss (L_{img}) is applied over the entire image area.

3.4 Asymmetric Geometry Preservation Mechanisms

In Phase 3, the background and dynamic subject Gaussians trained in the previous phases are merged into a single space and co-optimized. Since the auxiliary dataset contains no observations of the dynamic subjects, evaluating reconstruction losses on its views erases or distorts the subject geometry acquired from the main dataset. We therefore introduce a preservation mechanism that is activated only on training steps using auxiliary views. Its three components are complementary: opacity control suppresses asymmetric losses at the rendering level, while parameter freezing and density-control exclusion protect the subjects from residual updates driven by optimizer statistics and blind spots of the loss.

Opacity-Zero Rendering. Gaussians whose class probability exceeds a threshold, $p_i > \tau_{\text{cls}}$, are classified as dynamic subjects. For every auxiliary view, their opacities are set to zero ($\alpha_i = 0$) immediately before rasterization. The subjects thus neither occlude the auxiliary background nor receive reconstruction gradients from views in which they are absent.

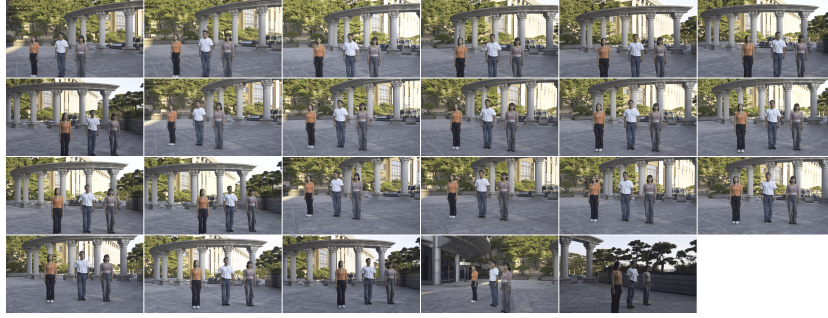
Gaussian Parameter Freeze. For the remaining gradient paths, all gradients of the subject Gaussian parameters (position μ_i , scale S_i , rotation R_i , opacity α_i , and intrinsic appearance feature sf_i) are masked to zero during auxiliary-view steps. Since zeroing the current gradients alone does not prevent parameter drift in optimizers with running statistics, we additionally reset the first and second moment estimates (m_t , v_t) of Adam [11] for the corresponding entries. The shared appearance and color networks are excluded from this freeze, allowing the background to adapt to the illumination of the auxiliary views.

No Densification & Pruning. Even with frozen parameters, adaptive density control can still alter the topology of the subject Gaussians through splitting, cloning, or pruning. We therefore exclude all Gaussians classified as dynamic subjects from densification and pruning on auxiliary-view steps, so that the subjects retain their geometric structure throughout the spatial expansion of the surrounding environment.

4 Experiments

4.1 Implementation details

All experiments were conducted in PyTorch on a single NVIDIA GeForce RTX 3090 GPU. We allocate 10k iterations to phases 1 and 2, respectively, and 20k iterations to phase 3, giving the background a total training budget of 30k iterations. In phase 3, the learning rates for the Gaussian positions, the appearance networks, and the sc_i coordinates are lowered to preserve the information established in the preceding phases, the class learning rate is set to zero, and densification and opacity resets are restricted to protect the previously formed Gaussian structures.



(a) Main set (23 images): captured around sunset by fixed cameras arranged in a grid formation, containing the primary target subjects.



(b) Auxiliary set (132 images): captured in the morning under a clear sky with a hand-held DSLR camera, covering the broader surrounding environment without the target subjects.

Fig. 3: Overview of our self-collected dataset. The two sets exhibit severe appearance discrepancies in illumination and camera characteristics, along with a structural disparity in the presence of the target subjects.

4.2 Dataset

Since no existing benchmark reflects our target scenario, in which a main capture containing core subjects must be spatially extended with auxiliary footage of the surrounding environment, we constructed a self-collected dataset of 155 images of an outdoor stage-like plaza featuring a colonnade structure. The main set consists of 23 images captured by fixed cameras arranged in a grid formation, simulating a practical multi-view stage capture setup. These images were taken around sunset and contain the primary target subjects standing on the plaza together with parts of the surrounding background. The auxiliary set consists of 132 images captured with a hand-held DSLR camera in the morning under a clear sky, covering a substantially wider area of the surrounding environment from diverse viewpoints. The auxiliary set was therefore captured at a different time of day with a different camera device, and the target subjects are entirely absent from it. As a result, the two sets exhibit severe appearance discrepancies arising

from the warm sunset illumination of the main views, the clear morning lighting of the auxiliary views, and the distinct color characteristics of the two camera sensors, as well as a structural disparity in which the target subjects appear exclusively in the main views. These properties faithfully reproduce the domain gaps of our target scenario, making the dataset well suited for evaluating core subject preservation. Representative samples of both sets are shown in Figure 3.

4.3 Qualitative Comparison

The qualitative evaluation results are shown in Figure 4.

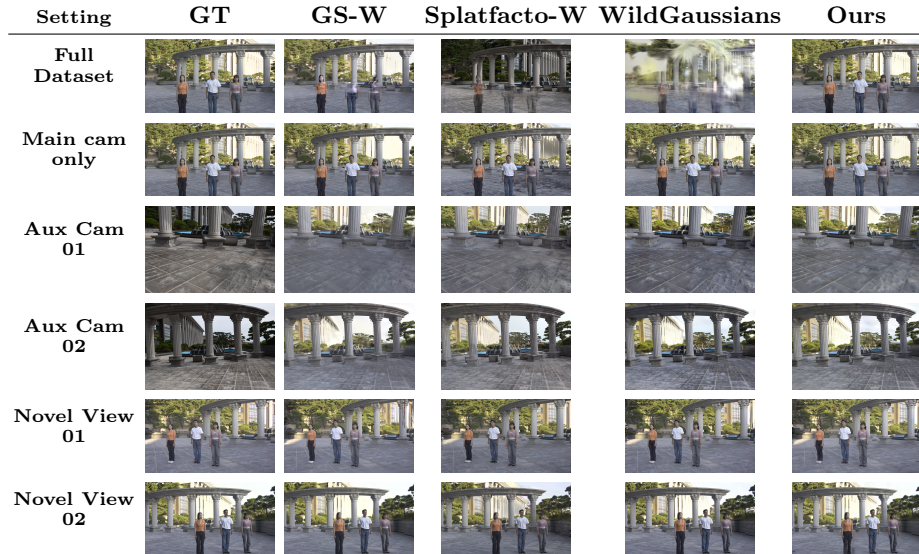


Fig. 4: Qualitative comparison across training configurations. In *Full Dataset*, the model trains on all views except one target main camera and is evaluated on that held-out view and *Aux Cam 01, 02* show auxiliary views color-matched to the main camera. In *Main cam only*, it trains solely on main-camera views without auxiliary supervision and *Novel View 01, 02* show newly rendered perspectives from unseen viewpoints.

In the whole-dataset training setup (**Full Dataset**, 155 images), baseline models failed to fully reconstruct the primary target object due to their transient occluder handling modules. Instead, they partially erased the object, introducing severe ghosting artifacts. Additionally, Splatfacto-w failed to accurately estimate the environmental conditions (e.g., lighting and time) corresponding to when the main camera images were captured, leading to color distortion. WildGaussians exhibited large and irregular artifacts across the entire scene. In contrast, thanks

to our proposed context-decoupled multi-phase framework, the proposed model faithfully reconstructed the primary target object near ground-truth quality.

In the main-camera-only training setup (**Main Cam Only**, 22 images), all comparison models achieved a generally high level of background alignment. However, due to the limited number of training views, structural collapse occurred in the 3D geometry of the background region located in the upper-right corner of the image. Specifically, GS-W suffered from overexposure in that background area due to strong lighting, failing to precisely reconstruct the structure, whereas Splatfacto-w rendered the same region dark, leading to a loss of texture. On the other hand, the proposed model maintained stable and natural background alignment even under sparse main-camera view conditions.

4.4 Quantitative Comparison

The quantitative evaluation results are presented in Table 1.

	Full Dataset			Main cam only		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
GS-W	<u>22.54</u>	0.8162	0.3199	21.97	0.8335	0.2826
Splatfacto-W-light	11.76	0.5921	0.4020	20.30	0.8624	0.1781
WildGaussians	12.41	0.5843	0.5919	<u>26.60</u>	0.8912	<u>0.1852</u>
Ha-NeRF	14.17	0.6991	0.4854	20.33	0.7665	0.4015
WonderJourney	11.67	0.3582	0.7393	11.67	0.3582	0.7393
HybridGS	11.53	0.5761	0.5490	-	-	-
DeSplat	20.20	<u>0.8367</u>	0.2394	-	-	-
DeGauss	16.80	0.6839	0.4076	-	-	-
Ours	26.99	0.8441	<u>0.2599</u>	26.87	<u>0.8634</u>	0.2167

Table 1: Quantitative comparison. *Full Dataset*: trained on all views except one held-out main-camera view used for evaluation. *Main cam only*: trained solely on main-camera views and evaluated identically. **Bold** and underline denote the best and second-best results. [†]In the *Main cam only* setting, HybridGS, DeSplat, and DeGauss do not perform their intended static/dynamic decomposition because no auxiliary-domain inconsistency is present; they reduce to conventional static-scene reconstruction. We therefore omit these redundant results from this diagnostic setting.

To comprehensively evaluate rendering quality, we adopt three standard metrics: PSNR to measure pixel-level color accuracy, SSIM to assess structural similarity, and LPIPS to evaluate perceptual realism consistent with human vision. Under the whole-dataset training setup (**Full Dataset**), the baseline models misidentified the primary target object which appears solely in the main camera views as a transient occluder, thereby failing to preserve its geometric structure. Consequently, they exhibited poor performance across overall evaluation metrics (PSNR, SSIM, and LPIPS). In contrast, the proposed model effectively protected

the information of the primary target object through the context-decoupled optimization framework and the asymmetric geometry preservation mechanism, achieving strong overall reconstruction performance.

Next, under the main-camera-only training setup (**Main Cam Only**), both the baseline models and the proposed model achieved high-quality background alignment within the limited views, recording overall strong quantitative scores.

It is noteworthy that the rendering quality of the proposed model trained on the full dataset (**Full Dataset**) surpassed that of the baseline models trained solely on the main camera images (**Main Cam Only**). Generally, jointly training on two datasets (main and auxiliary) captured at different times, under different lighting, and from distinct viewpoints introduces severe inconsistencies that easily lead to alignment failure. However, by leveraging a mechanism that physically decouples the background and the primary target object for stage-wise optimization, the proposed model effectively absorbed the 3D background geometry from the auxiliary dataset and successfully achieved high-quality alignment.

4.5 Ablation Studies

	Full		Core subject		Background	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
w/o multi-phase (single-stage)	21.23	0.7746	15.26	0.5534	23.52	0.8053
w/o Phase 2	25.49	0.8071	22.96	0.7383	25.99	0.8166
w/o all asym. mechanisms	22.80	0.7957	15.83	0.6418	26.29	0.8170
Ours (full)	26.99	0.8441	32.43	0.9288	26.38	0.8202

Table 2: Region-decomposed evaluation on the held-out main-camera view. *Core subject* and *Background* are computed over the semantic mask and its complement (core subject occupies 12.2% of pixels). The decomposition isolates the object-preservation effect that full-image metrics dilute.

We report full-image and region-level ablations in Tab. 2. Since the core subject occupies only 12.2% of the held-out view, we evaluate the subject and background separately using its semantic mask.

Removing the framework reduces full-image PSNR by 5.76 dB, but subject PSNR drops by 17.17 dB (32.43 $\rightarrow$ 15.26), compared with only 2.86 dB for the background. Without phase separation, auxiliary-view gradients directly modify subject Gaussians, causing artifacts and ghosting.

Ablating asymmetric preservation in Phase 3 reduces full-image PSNR by 4.19 dB and core-subject PSNR by **16.60 dB** (32.43 $\rightarrow$ 15.83), while the background remains nearly unchanged (26.38 $\rightarrow$ 26.29). Core-subject Gaussians also fall from 20,000 to 11,134, returning subject quality close to the single-stage result (15.26) and demonstrating the need to shield them from auxiliary-view updates.

Removing Phase 2 reduces full-image and core-subject PSNR by 1.50 and 9.47 dB, respectively, and leaves only 3,514 subject Gaussians. Thus, Phase 2 builds the subject representation, while asymmetric preservation maintains it; both are required to reach 32.43 dB.

	Class Threshold				
	0.1	0.3	0.5	0.7	0.9
PSNR $\uparrow$	25.80	25.41	26.99	25.88	26.35
SSIM $\uparrow$	0.8308	0.8318	0.8441	0.8327	0.8337
LPIPS $\downarrow$	0.2808	0.2776	0.2599	0.2763	0.2753

Table 3: Class Threshold Experiment on Full Dataset

We also conduct an ablation over the classification threshold (0.1–0.9). The default value of 0.5 achieves the best results across all metrics, while other thresholds show only marginal degradation, indicating the method is not highly sensitive to this hyperparameter.

5 Conclusion

We proposed a context-decoupled multi-phase framework that reconciles appearance differences between main and auxiliary captures while preserving core subjects. Experiments show that semantic class-based extraction and asymmetric Gaussian preservation incorporate auxiliary background details without degrading subject fidelity, enabling high-quality spatial extension.

6 Acknowledgements

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (2022-0-00022, RS-2022-II220022, Development of immersive video spatial computing technology for ultra-realistic metaverse services) (40%); the Culture, Sports and Tourism R&D Program through the Korea Creative Content Agency grant funded by the Ministry of Culture, Sports and Tourism in 2026 (Project Name: Multimodal Storyverse: A Multi-Agent Collaborative Platform for Expansive Narrative Creation, Project Number: RS-2026-25527029, Contribution Rate: 30%); and the Artificial Intelligence Innovation Graduate School (Sogang University) grant funded by the Korea government (MSIT) (RS-2026-25547954) (30%).

References

1. Barron, J.T., Mildenhall, B., Tancik, M., Hedman, P., Martin-Brualla, R., Srinivasan, P.P.: Mip-nerf: A multiscale representation for anti-aliasing neural radiance

- fields. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 5855–5864 (2021)
2. Barron, J.T., Mildenhall, B., Verbin, D., Srinivasan, P.P., Hedman, P.: Mip-nerf 360: Unbounded anti-aliased neural radiance fields. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 5470–5479 (2022)
 3. Carion, N., Gustafson, L., Hu, Y.T., Debnath, S., Hu, R., Suris, D., Ryali, C., Alwala, K.V., Khedr, H., Huang, A., et al.: Sam 3: Segment anything with concepts. arXiv preprint arXiv:2511.16719 (2025)
 4. Chen, X., Zhang, Q., Li, X., Chen, Y., Feng, Y., Wang, X., Wang, J.: Hallucinated neural radiance fields in the wild. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 12943–12952 (2022)
 5. Chung, J., Lee, S., Nam, H., Lee, J., Lee, K.M.: Luciddreamer: Domain-free generation of 3d gaussian splatting scenes. arXiv preprint arXiv:2311.13384 (2023)
 6. Dahmani, H., Bennehar, M., Piasco, N., Roldao, L., Tsishkou, D.: Swag: Splatting in the wild images with appearance-conditioned gaussians. In: European Conference on Computer Vision. pp. 325–340. Springer (2024)
 7. Fridman, R., Abecasis, A., Kasten, Y., Dekel, T.: Scenescape: Text-driven consistent scene generation. *Advances in Neural Information Processing Systems* **36**, 39897–39914 (2023)
 8. Fu, C., Zhang, Y., Yao, K., Chen, G., Xiong, Y., Huang, C., Cui, S., Cao, X.: Robustsplat: Decoupling densification and dynamics for transient-free 3dgs. In: 2025 IEEE/CVF International Conference on Computer Vision (ICCV). pp. 27126–27136. IEEE (2025)
 9. Huang, B., Yu, Z., Chen, A., Geiger, A., Gao, S.: 2d gaussian splatting for geometrically accurate radiance fields. In: ACM SIGGRAPH 2024 conference papers. pp. 1–11 (2024)
 10. Kerbl, B., Kopanas, G., Leimkühler, T., Drettakis, G., et al.: 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* **42**(4), 139–1 (2023)
 11. Kingma, D.P., Ba, J.: Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980 (2014)
 12. Kulhanek, J., Peng, S., Kukulova, Z., Pollefeys, M., Sattler, T.: Wildgaussians: 3d gaussian splatting in the wild. arXiv preprint arXiv:2407.08447 (2024)
 13. Martin-Brualla, R., Radwan, N., Sajjadi, M.S., Barron, J.T., Dosovitskiy, A., Duckworth, D.: Nerf in the wild: Neural radiance fields for unconstrained photo collections. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 7210–7219 (2021)
 14. Mildenhall, B., Srinivasan, P.P., Tancik, M., Barron, J.T., Ramamoorthi, R., Ng, R.: Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM* **65**(1), 99–106 (2021)
 15. Schonberger, J.L., Frahm, J.M.: Structure-from-motion revisited. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 4104–4113 (2016)
 16. Ungermann, P., Ettenhofer, A., Nießner, M., Roessle, B.: Robust 3d gaussian splatting for novel view synthesis in presence of distractors. In: DAGM German Conference on Pattern Recognition. pp. 153–167. Springer (2024)
 17. Wang, Y., Klasson, M., Turkulainen, M., Wang, S., Kannala, J., Solin, A.: Desplat: Decomposed gaussian splatting for distractor-free rendering. In: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 722–732. IEEE (2025)

18. Xu, C., Kerr, J., Kanazawa, A.: Splatfacto-w: A nerfstudio implementation of gaussian splatting for unconstrained photo collections. arXiv preprint arXiv:2407.12306 (2024)
19. Xu, J., Mei, Y., Patel, V.M.: Wild-gs: Real-time novel view synthesis from unconstrained photo collections. *Advances in Neural Information Processing Systems* **37**, 103334–103355 (2024)
20. Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., Zhao, H.: Depth anything v2. *Advances in Neural Information Processing Systems* **37**, 21875–21911 (2024)
21. Yang, Y., Zhang, S., Huang, Z., Zhang, Y., Tan, M.: Cross-ray neural radiance fields for novel-view synthesis from unconstrained image collections. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 15901–15911 (2023)
22. Ye, V., Li, R., Kerr, J., Turkulainen, M., Yi, B., Pan, Z., Seiskari, O., Ye, J., Hu, J., Tancik, M., et al.: gsplat: An open-source library for gaussian splatting. *Journal of Machine Learning Research* **26**(34), 1–17 (2025)
23. Yu, H.X., Duan, H., Hur, J., Sargent, K., Rubinstein, M., Freeman, W.T., Cole, F., Sun, D., Snavely, N., Wu, J., et al.: Wonderjourney: Going from anywhere to everywhere. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 6658–6667 (2024)
24. Yu, Z., Chen, A., Huang, B., Sattler, T., Geiger, A.: Mip-splatting: Alias-free 3d gaussian splatting. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 19447–19456 (2024)
25. Zhang, D., Wang, C., Wang, W., Li, P., Qin, M., Wang, H.: Gaussian in the wild: 3d gaussian splatting for unconstrained image collections. In: *European Conference on Computer Vision*. pp. 341–359. Springer (2024)
26. Zhang, K., Riegler, G., Snavely, N., Koltun, V.: Nerf++: Analyzing and improving neural radiance fields. arXiv preprint arXiv:2010.07492 (2020)